Anomaly Segmentation for Road Scenes: Pixel- vs Mask-based Post-hoc OoD Detection

Simone Paoletti Luca Camillini Lorenzo Ferrandi Emanuele Antonini
Politecnico di Torino
{s359956, s361268, s349309, s364176}@studenti.polito.it

Abstract

We study anomaly segmentation for autonomous driving by comparing the pixel-based ERFNet model with the mask-based EoMT architecture. We first evaluate EoMT checkpoints trained on different datasets through a shared protocol that accounts for their label-space mismatch, and then adapt the COCO-trained model to Cityscapes using three fine-tuning strategies. Progressive unfreezing achieves the best result, reaching 77.9% mIoU. For out-of-distribution detection, we compare MSP, MaxLogit, Max Entropy, and RbA across five road-scene benchmarks. MaxLogit is the most consistent general baseline, while RbA performs particularly well with the Cityscapes-trained EoMT. Temperature scaling improves softmax-based scores in some cases, but does not fully solve severe overconfidence. The code is available at https://github.com/Lucaaa31/Anomaly-Segmentation.

1. Introduction

Autonomous driving depends on accurate visual perception to act safely in traffic. Segmentation models give a vehicle a dense understanding of the road by labeling every pixel of the scene, yet they are trained under a closed-set assumption: all categories are fixed in advance. Real roads break it. An object never seen in training can appear, and a model that assigns it, with high confidence, to a known class endangers the vehicle. Anomaly segmentation addresses this gap by detecting and localizing the regions that fall outside the training distribution.

Scene understanding rests on a family of segmentation tasks. Semantic segmentation labels every pixel, as in the efficient ERFNet [13]; instance segmentation separates objects of the same class; and panoptic segmentation [9] unifies the two. Mask-based architectures reframe the problem as classifying a set of predicted masks: MaskFormer [3] introduced the idea, Mask2Former [4] added masked attention, and EoMT [8] shows that a plain DINOv2 [12] Vision Transformer can segment images without task-specific components. For the anomaly task, post-hoc scores read the outputs of a trained model to flag unknown regions [6, 11], and the SegmentMeIfYouCan [2] and Fishyscapes [1] benchmarks measure progress.

We evaluate these models for road-scene understanding along three axes. First, we compare two EoMT checkpoints, one trained on COCO [10] panoptic and one on Cityscapes [5] semantic segmentation, under a single protocol that maps both onto a common Cityscapes label space for a fair mIoU. Second, we adapt the COCO checkpoint to Cityscapes with three fine-tuning strategies, one of them based on LoRA [7]. Third, we contrast a pixel-based network (ERFNet) with the mask-based EoMT on anomaly segmentation, and measure how the scene representation affects the detection of unexpected objects.

2. Methodology

2.1. Setup and class-space mapping

We evaluate the two provided EoMT checkpoints on the 500 images of the Cityscapes validation split. Both models use the same DINOv2 ViT-Base/14 backbone: one was trained for semantic segmentation on Cityscapes with 19 classes, while the other was trained for panoptic segmentation on COCO with 133 categories. To obtain a consistent comparison, both checkpoints are evaluated on semantic segmentation using the same inference and evaluation pipeline. Performance is measured through mean Intersection over Union (mIoU). The same pipeline is later reused to evaluate the fine-tuned models of Section 2.2.

The two checkpoints cannot be compared directly because Cityscapes and COCO use different label spaces. The Cityscapes model already predicts the 19 target classes, whereas the COCO model predicts 133 categories. We align the two taxonomies through a fixed lookup table that maps COCO continuous IDs to Cityscapes train IDs whenever a meaningful correspondence exists.

The table covers the categories the two datasets share: *thing* classes (person, car, truck, bus, train, motorcycle, bicycle, traffic light) and *stuff* classes (road, sidewalk, building, sky, vegetation, terrain, fence, wall). Semantically related COCO labels are merged into the same target class, such as the wall variants mapped to `wall` and grass, dirt, and mountain mapped to `terrain`. Predictions without a valid Cityscapes correspondence are mapped to the `ignore` label and excluded from metric computation.

We report two label-space settings. The **strict** setting includes all 19 Cityscapes classes and therefore penalizes target categories that the COCO model cannot directly represent, such as `rider` and `pole`, while `traffic sign` can only be approximated through `stop sign`. The **common** setting uses a reduced space of 16 active classes supported by the adopted mapping: `pole` and `traffic sign` are ignored, while `rider` is merged into `person` in both predictions and ground truth. The strict setting evaluates the complete target taxonomy, whereas the common setting provides a fairer cross-dataset comparison by reducing the effect of incompatible class definitions.

2.2. Fine-tuning

The COCO checkpoint provides broad visual priors, but it was trained on a different label space and was not specialized for road scenes. We therefore adapt it to Cityscapes semantic segmentation. Compatible COCO weights are transferred to the target model, while the original 133-class prediction head and the 200-query embeddings are discarded because their dimensions do not match the 19-class, 100-query Cityscapes configuration. The pretrained DINOv2 ViT-Base/14 backbone with four register tokens and 3 EoMT mask-attention blocks are retained.

Common protocol. All strategies use the same optimization framework, with different step budgets and warm-up lengths. We train on the 2,975 Cityscapes training images using 640×640 crops, AdamW with weight decay 0.1, and mixed precision. A batch size of 2 with eight gradient-accumulation steps gives an effective batch size of 16. The learning rate is linearly warmed up and then reduced through cosine decay; the prediction head uses a peak learning rate of 10^{-4}, while the backbone uses 10^{-5} when trainable. Validation is performed every 250 steps on a fixed 100-image subset balanced across Frankfurt, Lindau, and Münster, while the final evaluation uses the complete 500-image validation split. Early stopping is applied after 12 validations without an improvement of at least 10^{-4}, except for the phased strategy, where temporary plateaus may occur before further layers are unfrozen.

Strategies. We compare three strategies that differ in which parameters they train (Table 1). `head_only_lora` freezes the backbone and trains the prediction head plus LoRA adapters (r=8, α=16) on the `qkv` and `proj` projections of every encoder block, about 7% of the parameters; it is the parameter-efficient baseline. `full` trains every parameter with a discriminative learning rate, 10^{-4} for the head and 10^{-5} for the backbone, so the pretrained features change slowly. `phased` unfreezes the backbone in five steps that expose its last $(0, 3, 6, 9, 12)$ blocks: phase 1 trains the head alone, phase 2 adds the three EoMT mask-attention blocks, and each later phase frees three more until the whole backbone trains in phase 5. Its optimizer holds all parameters from the start, so the backbone begins to move only once the cosine schedule is already decaying, which protects the pretrained encoder.

Strategy	Trainable	LoRA	LR (head/bb)	Steps
`head_only_lora`	head + LoRA	r=8	10^{-4} / –	12k
`phased`	head → all	–	10^{-4} / 10^{-5}	20k
`full`	all	–	10^{-4} / 10^{-5}	20k

Table 1. The three fine-tuning strategies. "bb" is the backbone; a dash means the backbone is frozen (LR 0). Warmup is 500/1000/750 steps respectively.

2.3. Post-hoc scoring

Pixel-based methods. Our first anomaly baselines are pixel-based. For pixel-based models like ERFNet, the network outputs a vector of raw logits per pixel. Here, we evaluate three post-hoc methods:

- **Maximum Softmax Probability (MSP):** the softmax converts the logits into class probabilities, and the anomaly score is computed as the complement of the largest probability. A pixel is considered more anomalous when the model is highly uncertain about its primary class prediction.
- **MaxLogit:** this method scores on the raw logits directly, keeping the magnitude and scale information that softmax normalizes away. The score is defined as the complement of the maximum logit value.
- **Maximum Entropy:** this method uses the entire probability distribution. A confused model spreads its probability mass across multiple classes, which increases the entropy. We normalize the entropy by the total number of classes and add a tiny epsilon factor for numerical stability.

Mask-based method. A mask-based architecture like EoMT does not output pixel logits directly. It groups pixels into binary mask regions and assigns class probabilities to each region, which opens a mask only post-hoc method:

- **Rejected by All (RbA):** this method reads the confidence of the proposed masks [11]. For each query, we take the maximum class probability, weight it by the spatial mask probability, take the maximum across all queries, and subtract this final confidence value from 1.0.

Evaluation metrics. We report two metrics on every dataset. The Area Under the Precision-Recall Curve (AuPRC) summarizes separability across thresholds. The False Positive Rate at 95% recall (FPR95) measures the share of in-distribution road pixels flagged as anomalies once the threshold catches 95% of the real anomalies.

2.4. Temperature scaling

Temperature scaling calibrates the logits before the softmax. It divides the logit vector by a single scalar temperature factor $T > 0$, which alters the resulting probability distribution. When $T > 1$ the distribution softens and the model's overconfidence drops; with $T < 1$ it sharpens and amplifies the winning class. We apply temperature scaling to MSP alone. Since T acts inside the softmax, MSP isolates the calibration effect cleanly. We used a fixed grid $T \in \{0.5, 0.75, 1.1\}$ against the uncalibrated baseline ($T = 1.0$). We also search a wider range for a T_{best}, chosen to maximize AuPRC and minimize FPR95.

3. Experimental Results

3.1. Fine-tuning

Following the protocol of Section 2.1, we evaluate the two provided EoMT checkpoints and the ERFNet baseline on the full Cityscapes validation. EoMT-CS reaches 81.7% mIoU. The pixel-based ERFNet baseline scores 72.0%, and EoMT-COCO trails both at 55.2%. The COCO model obtains 0% IoU for `rider` and `pole` and 4.7% for `traffic sign`, since these classes are missing or only partially represented in its original label space. This gap between EoMT-COCO and the two Cityscapes-aware models motivates the fine-tuning described in Section 2.2.

Training dynamics. Figure 1 compares the training loss and validation mIoU of the three fine-tuning strategies. `head_only_lora` reduces the training loss more rapidly, as it optimizes a smaller number of parameters, but reaches a lower validation plateau. `full` and `phased` fine-tuning achieve higher performance, while the `phased` strategy improves later because the backbone is progressively unfrozen. The best validation mIoU values are 76.5%, 78.3%, and 78.4% for `head_only_lora`, `full`, and `phased` fine-tuning, respectively.

Fine-tuning outcome. On the complete validation split, all three strategies substantially improve over the original COCO checkpoint. `Head_only_lora` reaches 75.7% mIoU, `full` fine-tuning obtains 76.9%, and `phased` fine-tuning achieves 77.9%. The largest improvements are observed for classes that the original model could not properly represent, such as `rider`, `pole` and `traffic sign`, as well as for less frequent classes such as `truck` and `train`. For the `phased` strategy, mIoU increases from 67.4% to 75.5%, 76.9%, 77.9%, and 78.4% across the five stages, on the subset of the evaluation of Cityscape. Nevertheless, the best fine-tuned checkpoint remains 3.8 percentage points below EoMT-CS.

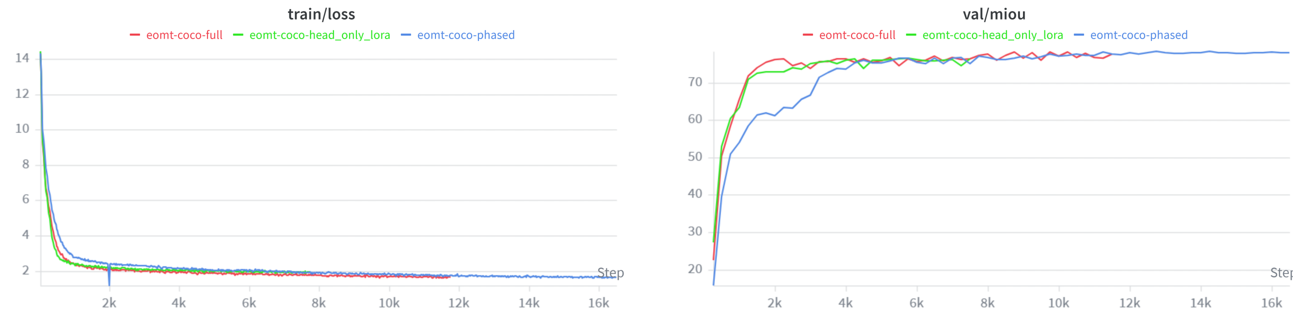
3.2. Anomaly detection

We compare the performance of the pixel-based baseline (ERFNet) against the mask-based architecture (EoMT) across five standard validation datasets. For each model, we test different post-hoc methods described in Section 2. The complete results are summarized in Table 2.

As shown in the table, the mean Intersection over Union for standard semantic segmentation stays exactly the same when we change the post-hoc method within the same checkpoint. This proves that performance on the known classes depends only on the frozen weights of the model, and not on the method we use to extract the anomaly score.

Quantitative evaluation matrix When analyzing the anomaly detection data, we can see clear trends for each architecture:

- **Pixel-based Baseline (ERFNet):** The MaxLogit method systematically gives the best results across all datasets. It significantly reduces the false positive rate (FPR95).
- **EoMT-COCO:** This configuration shows very different behaviors depending on the method. On one hand, MaxLogit achieves great peaks in AuPRC, reaching 80.83% on FS Static and 82.25% on Road Anomaly. On the other hand, controlling false positives is hard: on FS L&F all post-hoc methods exceed 96% FPR95, and the RbA method collapses on SMIYC RO-21, reaching an FPR95 of 94.78% while MSP and MaxLogit stay close to 12%. RbA only clearly helps on Road Anomaly, where it lowers the FPR95 to 74.36% against the 86–89% of the other methods.
- **EoMT-CS:** This is the most robust configuration in our tests. Traditional methods (MSP, MaxLogit, and Max Entropy) exhibit high stability across all datasets, with the sole exception of the AuPRC metric on SMIYC RO-21, which stays around 8%. On the other hand, the RbA method is far less stable: its FPR95 degrades strongly on SMIYC RA-21 (79.88%), RO-21 (93.73%), and FS Static (82.70%), even though its AuPRC remains comparable to that of the traditional methods. This suggests that RbA may not be the optimal approach for this specific scenario.
- **Fine-Tuned:** For both fine-tuned variants, performance is strongly dataset-dependent: the FPR95 saturates on



(a) training loss

(b) validation mIoU

Figure 1. Optimization dynamics of the three strategies (Weights & Biases logs): training loss and validation mIoU on the 100-image subset.

SMIYC RA-21 for all post-hoc methods, but drops to some of the best values in the table on FS Static and Road Anomaly. There does not seem to be a fixed pattern; however, it is worth noting that the Phased model with RbA achieves the lowest FPR95 on FS Static (5.83%) and on Road Anomaly (13.28%) across all configurations.

3.3. Temperature scaling

Looking at Table 3 that reports different temperature values, we can see clear trends depending on the dataset. Compressing the logits with smaller temperature values ($t = 0.5$ and $t = 0.75$) generally makes the performance worse. For example, on FS Lost&Found, dropping the temperature to $t = 0.5$ raises the FPR95 to 14.98% compared to 14.89% of the baseline. On the other hand, increasing the temperature just a little bit to $t = 1.1$ gives very conservative changes, and the results stay almost identical to the standard MSP baseline across most datasets.The most important trend happens at $t = 2$, which is the best configuration in this test. When we scale the logits with this parameter, the model gets a significant performance boost on almost all datasets. On SMIYC RA-21, the AuPRC increases from 74.30% to 77.82%, and the false positive rate (FPR95) drops from 33.43% to a much better 28.61%. It also achieves the highest AuPRC scores on FS Static (56.43%) and Road Anomaly (77.64%).However, even with the best calibration, this method still cannot solve the weak detection on SMIYC RO-21, where the AuPRC remains as low as 8.94% even at $t = 2$. This shows that while temperature scaling helps calibrate the probabilities, it cannot overcome the intrinsic difficulty of detecting the SMIYC RO-21 obstacles.

4. Discussion

4.1. Fine-tuning

The three strategies rank `phased` first, followed by `full` and `head_only_lora` (Sec. 3.1). Although the differences are small, they reflect the trade-off between adapting the COCO representations to road scenes and preserving the pretrained backbone.

Result analysis. `Head_only_lora` provides the most limited adaptation, since the backbone remains frozen and only about 7% of the parameters are updated; its shorter 12k-step schedule may also affect the result. `full` fine-tuning adapts the entire model but remains about one point below phased fine-tuning. The `phased` strategy updates the model more gradually, first training the prediction components and then progressively unfreezing the backbone as the learning rate decreases.

The largest improvement occurs between P1 and P2, where mIoU rises from 67.4% to 75.5% after unfreezing the last three query-processing blocks. Later phases provide smaller gains, suggesting that most of the adaptation occurs in the upper part of the network. All fine-tuned models remain below EoMT-CS, which was trained directly on Cityscapes, while our experiments were limited to 640×640 crops and a restricted compute budget.

Methodological critique. The experiments use a single seed, so the small gap between `phased` and `full` fine-tuning may be affected by run-to-run variability. The training schedules also differ across strategies. `full` and `head_only_lora` used early stopping, which halted training after 2000 steps (about 11 epochs) without improvement on the validation subset, whereas `phased` disabled it so that all five phases could complete. This, together with the shorter schedule of `head_only_lora`, makes it difficult to separate the effects of parameter freezing from training duration. Checkpoints were selected on a fixed 100-image validation subset, and LoRA was only tested with a frozen backbone, so its individual contribution cannot be isolated.

Table 2. Anomaly segmentation performance comparison between pixel-based (ERFNet) and mask-based (EoMT) architectures using different post-hoc methods across five validation datasets.

Model	mIoU (%)	Method	SMIYC RA-21 AuPRC	SMIYC RA-21 FPR95	SMIYC RO-21 AuPRC	SMIYC RO-21 FPR95	FS L&F AuPRC	FS L&F FPR95	FS Static AuPRC	FS Static FPR95	Road Anomaly AuPRC	Road Anomaly FPR95
ERFNet	71.99	MSP	29.10	62.55	2.71	65.22	1.75	50.59	7.47	41.84	12.42	82.58
		MaxLogit	38.32	59.34	4.63	48.44	3.30	45.49	9.50	40.30	15.58	73.25
		Max Entropy	30.97	62.66	3.04	65.91	2.58	50.16	8.84	41.55	12.67	82.75
EoMT-COCO	55.16	MSP	56.89	39.54	74.60	11.99	1.32	96.60	80.09	16.66	82.24	89.04
		MaxLogit	58.59	32.31	**74.63**	12.44	1.24	96.64	80.83	15.09	**82.25**	87.96
		Max Entropy	59.70	**25.59**	73.82	42.41	0.99	96.66	82.15	14.06	82.24	86.76
		RbA	51.09	59.45	72.05	94.78	0.32	97.82	**84.67**	46.56	80.81	74.36
EoMT-CITYSCAPES	81.68	MSP	74.30	33.43	7.90	16.59	25.42	14.89	54.32	35.20	75.68	19.33
		MaxLogit	73.87	33.55	7.93	16.74	25.44	14.64	54.46	34.84	75.15	19.08
		Max Entropy	**76.76**	33.31	8.86	17.64	27.96	14.70	52.68	35.12	77.62	18.82
		RbA	69.88	79.88	8.17	93.73	25.97	**8.76**	56.09	82.70	74.46	20.40
EoMT-FT-phased	77.91	MSP	63.39	99.97	12.69	6.71	43.59	21.65	78.14	11.27	73.47	18.85
		MaxLogit	63.12	99.97	12.76	6.46	43.76	21.17	78.55	11.02	73.25	18.45
		Max Entropy	63.41	99.97	11.69	6.04	41.80	21.21	79.02	10.97	73.66	18.01
		RbA	57.03	99.20	12.47	69.99	**44.99**	42.13	82.24	**5.83**	71.39	**13.28**
EoMT-FT-full	76.85	MSP	53.23	98.19	10.68	7.73	41.93	16.60	71.83	13.33	60.03	17.29
		MaxLogit	53.05	98.11	10.66	7.39	42.12	16.01	72.35	13.04	59.73	17.11
		Max Entropy	52.85	96.27	11.05	6.77	41.52	15.74	73.09	12.76	60.47	16.74
		RbA	41.21	92.89	10.33	**5.44**	43.00	27.04	75.56	8.29	57.51	16.44

Table 3. Anomaly segmentation performance with temperature calibration using MSP across five validation datasets.

Model	mIoU (%)	Method	SMIYC RA-21 AuPRC	SMIYC RA-21 FPR95	SMIYC RO-21 AuPRC	SMIYC RO-21 FPR95	FS L&F AuPRC	FS L&F FPR95	FS Static AuPRC	FS Static FPR95	Road Anomaly AuPRC	Road Anomaly FPR95
EoMT-CITYSCAPES	81.68	MSP ($t = 1$)	74.30	33.43	7.90	16.59	25.42	14.89	54.32	35.20	75.68	19.33
		MSP ($t = 0.5$)	74.11	33.05	7.71	16.94	25.33	14.98	54.18	35.12	75.44	19.21
		MSP ($t = 0.75$)	75.03	31.88	7.49	17.61	24.71	15.52	53.41	36.41	76.12	18.44
		MSP ($t = 1.1$)	73.76	34.12	8.21	16.08	25.19	15.07	54.77	34.55	75.29	19.88
		MSP (best $t = 2$)	**77.82**	**28.61**	**8.94**	**14.77**	**27.31**	**13.18**	**56.43**	**32.07**	**77.64**	**16.82**

4.2. Anomaly detection

Generalization vs Specialization The choice of datasets for pre-training and fine-tuning shows a clear tension between general open-world knowledge and specific domain specialization: the EoMT-COCO model has never seen specific driving environments, which is why its standard segmentation score is low (55.16% mIoU). However, because COCO contains 80 different real-world object classes, the model learns highly generalized shapes. When using MaxLogit, this general knowledge helps it spot unusual objects easily, leading to high AuPRC scores like 80.83% on FS Static. Fine-tuning the model on road scenes adapts the features to the driving domain, but it also creates an extreme "closed-world" rigidity. This overfitting does not raise the FPR95 uniformly: it collapses almost entirely on SMIYC RA-21, while remaining competitive on the other four benchmarks. In this rigid setup, the RbA method still provides the best local trade-offs (such as 5.83% FPR95 on FS Static for the phased model) because it relies on the geometric imperfections of the masks rather than the overfitted semantic features.

4.3. Temperature scaling

As we mentioned in the Section 3.3, the temperature scaling experiments show that the best parameter is $t = 2$. Since $t > 1$, this setup helps soften the network's confidence scores. By smoothing out these sharp logit distributions, a higher temperature helps the post-hoc method find the true uncertainty that was previously hidden under those high probabilities. This explains why we see a clear performance boost in both AuPRC and FPR95 across almost all datasets. However, temperature scaling cannot resolve severe detection failures. This is clearly proven by the behaviour on the SMIYC RO-21 dataset, where the AuPRC stays as low as 8.94% even with the best calibration.

5. Conclusion

Our results show that adapting the COCO-trained EoMT model to CS substantially improves semantic segmentation, with phased fine-tuning achieving the best adaptation. Higher performance on known classes does not directly translate into better anomaly detection: OoD performance strongly depends on the training domain, the segmentation architecture, and the adopted scoring method. MaxLogit provides a reliable pixel-based baseline, while RbA yields the best local FPR95 trade-offs on the fine-tuned models. Temperature scaling improves confidence-based detection in several cases, but cannot overcome the weak detection seen on SMIYC RO-21.

Acknowledgments

The authors wrote this report without any generative AI tool. A generative AI tool supported the codebase only. **System and version:** Claude (Opus 4 model family), accessed through the Claude Code assistant. **Publisher:** Anthropic. **Context and use:** the tool helped debug the Colab notebooks, restructure the repository, and implement parts of the evaluation and fine-tuning pipelines.

The authors reviewed, tested, and edited every AI-generated output. All engineering and experimental decisions, including the fine-tuning strategies, the post-hoc scoring methods, and the interpretation of the results, were made and validated by the authors.

Revision Notes

Following the review process, we have carefully addressed the issues raised regarding our report: the error in the key acronym definition and the unexpected anomaly segmentation results.

First, regarding the terminology, we acknowledge that a human error and a conceptual misunderstanding led to the incorrect definition of the RbA acronym in the original report. This has now been corrected according to the paper [11], changing the definition from “Region-based Anomaly detection” to “Segmenting Unknown Regions Rejected by All”. In accordance with this correction, we systematically reviewed and updated our notebook pipeline to ensure adherence to the proper mathematical methodology.

Second, regarding the experimental framework and the resulting metrics, the pipeline correction naturally led to new RbA results that differ significantly from the previous version. Furthermore, we identified that executing the experimental notebooks individually and sequentially in our initial setup could introduce inconsistencies. To permanently resolve this, we carried out a complete, and clean re-run of the entire experimental within a single, continuous execution block. This process brings about slightly different results, which have been updated accordingly across all tables and comments.

References

[1] Hermann Blum, Paul-Edouard Sarlin, Juan Nieto, Roland Siegwart, and Cesar Cadena. The Fishyscapes benchmark: Measuring blind spots in semantic segmentation. *IJCV*, 129 (11):3119–3135, 2021. 1

[2] Robin Chan, Krzysztof Lis, Svenja Uhlemeyer, Hermann Blum, Sina Honari, Roland Siegwart, Pascal Fua, Mathieu Salzmann, and Matthias Rottmann. SegmentMeIfYouCan: A benchmark for anomaly segmentation. In *NeurIPS*, 2021. Datasets and Benchmarks Track. 1

[3] Bowen Cheng, Alexander G. Schwing, and Alexander Kirillov. Per-pixel classification is not all you need for semantic segmentation. In *NeurIPS*, 2021. 1

[4] Bowen Cheng, Ishan Misra, Alexander G. Schwing, Alexander Kirillov, and Rohit Girdhar. Masked-attention mask transformer for universal image segmentation. In *CVPR*, 2022. 1

[5] Marius Cordts, Mohamed Omran, Sebastian Ramos, Timo Rehfeld, Markus Enzweiler, Rodrigo Benenson, Uwe Franke, Stefan Roth, and Bernt Schiele. The Cityscapes dataset for semantic urban scene understanding. In *CVPR*, 2016. 1

[6] Dan Hendrycks, Steven Basart, Mantas Mazeika, Andy Zou, Joseph Kwon, Mohammadreza Mostajabi, Jacob Steinhardt, and Dawn Song. Scaling out-of-distribution detection for real-world settings. In *ICML*, 2022. 1

[7] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *ICLR*, 2022. 1

[8] Tommie Kerssies, Niccolò Cavagnero, Alexander Hermans, Narges Norouzi, Giuseppe Averta, Bastian Leibe, Gijs Dubbelman, and Daan de Geus. Your ViT is secretly an image segmentation model. In *CVPR*, 2025. 1

[9] Alexander Kirillov, Kaiming He, Ross Girshick, Carsten Rother, and Piotr Dollár. Panoptic segmentation. In *CVPR*, 2019. 1

[10] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO: Common objects in context. In *ECCV*, 2014. 1

[11] Nazir Nayal, Mısra Yavuz, João F. Henriques, and Fatma Güney. RbA: Segmenting unknown regions rejected by all. In *ICCV*, 2023. 1, 2, 6

[12] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, et al. DINOv2: Learning robust visual features without supervision. *Trans. Mach. Learn. Res. (TMLR)*, 2024. 1

[13] Eduardo Romera, José M. Álvarez, Luis M. Bergasa, and Roberto Arroyo. ERFNet: Efficient residual factorized ConvNet for real-time semantic segmentation. *IEEE Trans. Intell. Transp. Syst.*, 19(1):263–272, 2018. 1